OPEN XDR: EXPLAINED

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L01 - INTRODUCTION AND AGENDA

L01A-PRES: Open XDR Explained Intro Agenda (02m:18s)

00:00-01:00
Welcome to the *Open XDR: Explained* module of the SOC Analyst course. In this module we'll be exploring concepts related to understanding what Open XDR is, what benefits it conveys, and what differentiates Open XDR from other security platforms. Later in the course we'll learn about the Stellar Cyber user interface and how to conduct investigations. This course, however, is intended to give a high-level view of what Stellar Cyber Open XDR is and what security problems it solves for SOC analysts.

This module's lessons include:

- *What Is Open XDR*? This lesson describes what Open XDR represents. Open XDR is a unique type of security product. In this lesson, we'll explore what Open XDR is and how it works.

- *Benefits of Open XDR*, which explores the value added by deploying Stellar Cyber Open XDR in your Information Security Architecture.

01:00-02:00

- *Features of Open XDR* speaks to the unique security benefits of using Stellar Cyber.

- *Components of Open XDR* takes a look at the mandatory and optional data sources that power protection in Open XDR. It explains how the layered coverage provides unique visibility into your network's security.

- The *Modular Sensor Functions* lesson shows how modular sensors drive powerful detection-and-response capabilities to achieve your expected security outcomes.

- The lesson, *Threat Detection Enrichment* looks at how enrichment of security data allows analysts to understand security events, attacks, breaches, abnormal processes, and identity threats.

- The *Response Actions* lesson explores how API connections to third-party platforms enable a range of actions that can be automated or taken manually, as well as be integrated into playbooks.

02:00-02:18

- And lastly, the *Key Differentiators of Open XDR* lesson demonstrates how Stellar Cyber provides unique security, identity, and infrastructure protection through its unique integration of capabilities.

L02 - WHAT IS OPEN XDR?

L02A-PRES: What Is Open XDR? (06m:10s)

00:00-01:00

The *Open XDR: Explained* module will start with the lesson, *What Is Open XDR? What Is Open XDR?* Open XDR is a platform for extended detection and response that ingests security Alerts, logs, and telemetry from any security vendor and combines them with its own network detection and response that includes an intrusion-detection system, deep packet inspection, threat intelligence, network detection and response, OT security, Case management, a malware sandbox, an automated response, all in one platform. Additionally, custom content can be built in Stellar Cyber to detect threats specific to your environment. Cases can be investigated in a timeline view and augmented with a graphic display of affected users and assets.

01:00-02:00
We'll go deeper into the types of integration supported by Stellar Cyber later in this module, but here's an overview of how the platform works.
Essentially, from left to right we have aggregation of many security data types, collection, normalization, storage, machine learning, alerting, threat hunting, automatic correlation, response actions, Case management, and reporting.

Tools and telemetry consists of logs, analytics, Alerts, and security events from Stellar Cyber and third-party platforms.

Collect is where data gets ingested, parsed, normalized, enriched, and stored.

Detect consists of machine-learning Alerts, rule-based Alerts, and threat hunting.

There are two types of machine learning in Stellar Cyber. Unsupervised machine learning models power the Alert types that look for dissimilarities from normal patterns.

02:00-03:00
These Alert types trigger when activity appears significantly different from the observed normal pattern. An example of this type of rule is account MFA login failure anomaly.

Supervised machine-learning models power the Alert types that look for known bad patterns based on training performed on large-scale data sets. Supervised machine-learning models require no training period because they are already trained and ready to be used. In summary, Supervised machine learning is used when there are clear identifiers for what bad looks like, but it is far too computationally complex for a simple rule to be used. An example is a DGA rule. That is, a host is using a potential domain-generation algorithm, or DGA. If the target domain is a malicious domain, the host might be compromised. Investigate the DGA domains and the host.

03:00-04:00

Rule-based Alerts are Alerts that are based on certain logical criteria being met to trigger an Alert.

Threat hunting involves actively searching for cyber threats. It is a proactive approach to cybersecurity that goes beyond traditional security measures. Stellar Cyber provides various tools and features for threat hunting including automated threat hunting, dashboards, and response actions. By leveraging machine learning and analytics, Stellar Cyber helps analysts identify and investigate potential security threats in real time. Threat hunting in Stellar Cyber involves analyzing raw event data, examining anomalies, and generating Alerts to prioritize actions and responses.

Correlated Alerts is a game-changing feature in Stellar Cyber. Cases will be explored in working with Cases and Alerts later on, but how it works is Cases are formed through an automated process that leverages machine learning to correlate disparate Alerts into cohesive Cases.

04:00-05:00
A Case in Stellar Cyber is essentially a set of multiple correlated Alerts and entities that together constitute a potential unified security threat. The system dynamically ranks these Cases by a score indicating the severity of the attack. This is a huge time saver and lays out the narrative of events neatly and chronologically. It cuts investigation time down dramatically and allows a much more rapid response.

Investigate and respond is the portion of the workflow in Stellar Cyber where a threat has been identified and action is required. Automated or manual response actions are usually the most immediate action to be taken in the event of activity that needs further investigation. We'll discuss response actions and their impact on security operations later in this module.

Recommended remediations come in the form of recommendations in the user interface based on which connectors are seeing the specific data. Stellar Cyber automatically recommends actions to you based on the Case's specifics and the connectors providing data for the observables in the Case.

05:00-06:10
For example, if a firewall provided the data for a host in your Case, the **Response** tab typically includes a block IP action for that host.

Case management is an essential part of investigating and responding to threats.

Sharing knowledge between teams and information security and stakeholders that have an input or make decisions about operations is crucial. At times the sharing of this knowledge needs to be immediate. Stellar Cyber's reporting capabilities include the generation of comprehensive and customizable reports that detail security incidents, Alerts, and overall system health. These reports can be tailored to specific requirements providing insights into threat patterns, compliance with security policies, and operational efficiency. This functionality supports informed decision making and demonstrates compliance with regulatory, enhancing the security posture and facilitating communication across an organization.

The lesson, *What Is Open XDR*, is complete. You should now be familiar with Open XDR. Thank you!

L03 - BENEFITS OF OPEN XDR

L03A-PRES: Benefits of Open XDR (03m:45s)

00:00-01:00
The next lesson in the *Open XDR: Explained* module is the *Benefits of Open XDR*.
Benefits of Open XDR. The main benefits of the Stellar Cyber Open XDR platform include enhanced visibility across the network, endpoints, and cloud environments, enabling comprehensive threat detection and response.

Stellar Cyber integrates data from various security tools, into a single platform, facilitating advanced analytics and automated incident response.

Data integration streamlines security operations, reduces complexity, and improves efficiency, offering a proactive stance against cyber threats.

Presenting all events in an easily navigable format enables a deeper understanding of each Alert and event. This approach also breaks down the isolated data compartments that are often created by individual security tools.

01:00-02:00
Stellar Cyber empowers analysts to see a holistic view of activity, instead of having to view network and asset activity in individual tools.

As an analyst, I've done it both ways. In Stellar Cyber, the automated creation of Cases provides a complete visual overview of the entire event, unlike the limited view offered by individual tools. Additionally, having detailed descriptions and research tools built directly into Alerts help analysts determine the right response action quickly.

Faster mean time to response: Having all your events in one view and having clear, thorough descriptions of the nature of the activity allow for an attack narrative to reveal itself instantly. Stellar Cyber reduces Mean Time to Response (MTTR) by automating the detection, analysis, and response processes within its Open XDR platform.

It leverages advanced analytics, machine learning, and integrated security tools to quickly identify and prioritize threats.

02:00-03:00
Automated workflows and pre-defined incident response actions enable rapid containment and remediation of threats, significantly cutting down the time security teams spend on manual tasks and accelerating the overall response to cyber incidents.

Automated response: Save countless analyst work hours by automating responses to threats. The best way to respond to an incident is to prevent it in the first place.

Automatic incident correlation: Stellar Cyber correlates the key elements of data you need to quickly respond to an event. Native, third-party, and custom Alerts are all tied together by proprietary technology, allowing an analyst to connect the dots that otherwise might not get connected. It is a tool that is designed to collect and respond through the entire security stack.

Let's take a moment to check your knowledge.

What is Open XDR? Review, and then select the best option.

A. An open security platform for extended detection and response
B. An email security platform
C. A cloud computing service
D. A data normalization tool

03:00-03:45
The correct answer A. Open XDR is an open security platform for extended detection and response. It improves threat detection and response by consolidating and integrating disparate security tools.

What is a benefit of Open XDR's integrated view?

A. Slower mean time to response
B. Manual incident correlation
C. Automated response
D. One single vendor log to review

The correct answer is C; a benefit of Open XDR's integrated view is automated response.

The lesson, *Benefits of Open XDR*, is complete. You should now be familiar with the unique security benefits of Open XDR. Thank you!

L04 - FEATURES OF OPEN XDR

L04A-PRES: Features of Open XDR (03m:45s)

00:00-01:00
The next lesson in the *Open XDR: Explained* module is *Features of Open XDR*. Features of Open XDR speaks to the unique security benefits of using Stellar Cyber.

- **Integrations**: An approach to Extended Detection and Response that leverages open integrations with various security tools, regardless of vendor.
- **Vendor Agnostic**: Unlike proprietary XDR solutions, Open XDR works across different vendors' products, offering flexibility in security stack composition.
- **Integration and Interoperability**: Facilitates seamless integration and interoperability between different security technologies and platforms.
- **Customization and Control**: Offers organizations the ability to customize their security setup by choosing best-of-breed components.
- **Unified Management**: Provides a centralized management console for monitoring and responding to threats across diverse security solutions.

01:00-02:00

- **Data Source Inclusivity**: Capable of ingesting data from a wide range of sources, including third-party tools and cloud services.
- **Secure Investments**: Reduces the need for complete overhauls of existing security infrastructure, protecting previous investments. Adds flexibility from outright dependencies other vendors create with their closed products.
- **Scalability**: Easily scales with organizational needs without being locked into a single vendor's ecosystem.
- **Community-Driven Enhancements**: Benefits from community-driven developments and shared threat intelligence.
- **Flexibility in Response Actions**: Enables flexible response actions utilizing a variety of tools from different vendors within the security stack.
- **Comprehensive Coverage**: Aims to provide comprehensive threat detection and response across the entire technology stack, including endpoints, networks, and cloud.

02:00-03:00

- **AI and Machine Learning**: It harnesses the power of artificial intelligence and machine learning from diverse systems to enhance threat detection and predictive analysis.
- **Real-Time Collaboration**: It facilitates real-time collaboration and information sharing between different security components for faster threat resolution.
- **Overcomes difficult security challenges**: It facilitates compatibility and complexity in managing disparate systems while ensuring cohesive security operations.

Let's take a moment to check your knowledge.

What does "vendor agnostic" mean in the context of Open XDR?

1. It works only with one specific vendor's products.
2. It doesn't work with any vendor's products.
3. It works with various security vendors' products.
4. It is not related to vendors.

The correct answer is C; in the context of Open XDR, "vendor agnostic" means that it works with various security vendors' products.

03:00-03:45

True or false? Open XDR uses the power of AI and machine learning from diverse systems to enhance threat detection and predictive analytics.

The correct answer is True: Open XDR harnesses the power of Artificial Intelligence and machine learning from diverse systems to enhance threat detection and predictive analytics.

The lesson, *Features of Open XDR*, is complete. You should now be familiar with the unique features of Open XDR. Thank you!

L05 - COMPONENTS OF OPEN XDR

L05A-PRES: Components of Open XDR (18m:30s)

00:00-01:00
The next lesson in the *Open XDR: Explained* module is *Components of Open XDR*. This lesson, *Components of Open XDR*, is where we get into the real power of the product. Now that we've discussed features and benefits, we'll take a look at the powerful integrations and native components Stellar Cyber uses to protect the network. We'll go over each component in case any of the concepts are new to you. Let's get started!

The integrations are what make this tool such a great way to manage your security operations. Stellar Cyber leverages third-party integrations by incorporating a wide range of security tools and data sources into its Open XDR platform to provide comprehensive security operations. These integrations enable the aggregation, normalization, and analysis of data across different layers of an organization's security posture, enhancing visibility and threat detection.

01:00-02:00
By utilizing external intelligence and response capabilities, Stellar Cyber streamlines workflows, automates responses, and ensures a more effective and coordinated defense against cyber threats. Additionally, Stellar Cyber's native security capabilities and native data management and analysis tools add additional layers of coverage to achieve your desired security, identity, and infrastructure outcomes. Some of the third-party integrations that drive detection and response are endpoint detection and response, or EDR. These products are designed to monitor endpoint and network events while storing this information in a centralized database where further analysis, detection, investigation, reporting, and alerting takes place. They provide advanced threat detection, investigation, and response capabilities to identify and mitigate cyber threats such as malware, ransomware, and the activities of malicious actors.

02:00-03:00
EDR solutions offer tools for forensic analysis, allowing organizations to investigate incidents post breach and improve their security posture against future threats.

Identity and Access Management (or IAM) products **manage** digital identities and their access to various systems and environments within an organization. They **ensure** that the right individuals gain access to the right resources at the right times for the right reasons, enhancing organizational security and compliance. IAM systems **facilitate** the automation of user provisioning, role-based access control, authentication, and single sign-on (or SSO) services, streamlining the management of user identities and permissions across a wide range of applications and systems.

Stellar Cyber integrates with a variety of **cloud security tools** to **enhance cybersecurity across different environments.**

03:00-04:00
Cloud Security Posture Management is a very important area for protection in today's IT environments. Just a few years ago, many organizations that worked with sensitive data were hesitant to move their operations to the cloud. Today, it's almost impossible NOT to have data in the cloud. Some of these cloud security vendors are Amazon Web Services (or AWS), Microsoft Azure, Google Cloud Platform, and Oracle Cloud Infrastructure.
This extensive integration capability allows organizations to maintain control while choosing the cloud security posture management (or CSPM) products that best fit their needs, ensuring consistent security outcomes across all their cloud environments.

Email security tools protect against attacks like phishing. According to one report, 79% of takeover attacks start with phishing. These tools often look for malicious code or attachments in emails. They are an important area that needs protection. Some examples of email tools that integrate with Stellar Cyber are Proofpoint, Broadcom (or Symantec), Barracuda Networks, and Mimecast.

04:00-05:00
These integrations allow Stellar Cyber to provide a more unified and comprehensive security posture against the myriad email-based threats that organizations face today.

Secure Access Service Edge (or SASE) is a cybersecurity framework that combines comprehensive wide-area networking (or WAN) capabilities with cloud-native security functions.
It aims to deliver secure access to cloud-based services and applications by integrating network security services like secure web gateways (or SWGs), cloud access security brokers (CASB), firewall as a service (FWaaS), and zero-trust network access (ZTNA) within a single, unified cloud service.
SASE supports the dynamic, secure access needs of organizations' distributed workforces, offering simplified management, improved security posture, and reduced complexity by merging network and security services.

05:00-06:00
Software as a Service (SaaS) in cybersecurity refers to cloud-based applications delivered over the internet that provide users with security services to protect against threats and
Manage security protocols. These services range from email protection and identity management to threat detection and response, enabling organizations to deploy advanced security measures without the need for on-premises hardware. SaaS cybersecurity solutions offer scalability, ease of use, and continuous updates, allowing businesses to stay ahead of evolving cyber threats while reducing the complexity and cost of traditional security infrastructure.

Let's take a moment to check your knowledge.

True or false? Third-party integrations enhance visibility and threat detection.

The correct answer is True: Third-party integrations enhance visibility and threat detection by enabling the aggregation, normalization, and analysis of data across different layers of an organization's security posture.

06:00-07:00
Stellar Cyber leverages third-party integrations to:

A. Provide comprehensive security operations
B. Streamline workflows
C. Automate responses
D. Ensure a more effective defense against cyber threats
E. All the above

The correct answer is E: all the above.

Native Data Management & Analysis is a majorly impactful aspect of Stellar Cyber's protection. This slide may not do it justice because so much of the coverage comes from Modular Sensors and Server Sensors, but we'll cover all that protection later in this module.
The Stellar Cyber API Integration Engine allows for programmatic interaction with the Stellar Cyber platform. It enables users to write scripts to retrieve data or create entities within the system, supporting a range of functionalities such as managing connectors, retrieving sensor and Case information, updating Case details, and managing tenants and user activities.

07:00-08:00
This flexibility facilitates the automation and customization of security operations within the Stellar Cyber ecosystem.
Stellar Cyber's Detection Engine is a core component of its cybersecurity platform, designed to automatically identify and alert on potential security threats across an organization's digital infrastructure. Utilizing advanced analytics, machine learning, and integration of diverse security data sources, the engine provides comprehensive threat-detection capabilities. It analyzes network traffic, logs, and other data to detect anomalies, malicious activities, and indicators of compromise, helping Helps security teams to quickly respond to and mitigate potential security incidents.
Stellar Cyber's Correlation Engine uses advanced analytics and machine learning to identify and correlate security events across various data sources, enhancing threat detection and reducing false positives.

08:00-09:00
It aggregates and analyzes data from the network, endpoint, and cloud sources, applying security rules and behavioral analysis to pinpoint real threats. This process enables the engine to provide actionable insights and prioritize alerts, helping security teams focus on the most critical issues efficiently. It creates Cases that vastly improve the way analysts interact with cyber alerts and events.

Stellar Cyber's Data Lake is a centralized repository. It is designed to store, manage, and analyze large volumes of disparate security data from across an organization's entire digital environment.

This enables the consolidation of data from network traffic, logs, endpoints, and cloud sources, Facilitates advanced threat detection, investigation, and response. The Data Lake supports scalable data analysis, helping security teams to efficiently process and leverage security data for improved visibility and actionable insights into their security posture.

09:00-10:00

This enables comprehensive data analysis and advanced threat detection by providing a single point of access to vast amounts of security-related information. It supports the processing and analysis of data at scale, allowing for efficient investigation, detection, and response to cyber threats. The Data Lake is designed to handle high volumes of data, making it a foundational element of Stellar Cyber's security operations platform for delivering actionable intelligence.

Let's take a moment to check your knowledge.

Which component creates Cases?

A. The API Integration engine
B. The Detection engine
C. The Correlation engine
D. The Sensor engine

The correct answer is C. The Correlation engine creates Cases, which vastly improve the way analysts interact with cyber alerts and events.

10:00-11:00

How does Stellar Cyber's Open XDR platform handle data from various sources?

A. It discards all non-essential data.
B. It only analyzes data from third-party tools.
C. It normalizes data into multiple formats.
D. It normalizes data into a single format, stores all data in a Data Lake, and analyzes it.

The correct answer is D; Stellar Cyber's Open XDR platform normalizes data into a single format, stores all data in a Data Lake, and analyzes it.

Stellar Cyber's Network Detection and Response (or NDR) capabilities leverage advanced analytics and machine learning to monitor network traffic in real-time, detecting and responding to threats before they can impact the organization. It provides comprehensive visibility across the network, identifying suspicious activities and anomalies that indicate potential security incidents. With automated response options, it enables quick containment and remediation of threats, enhancing the overall security posture and resilience against cyber attacks.

11:00-12:00

Stellar Cyber's User and Entity Behavior Analytics (or UEBA) capabilities focus on detecting insider threats, compromised accounts, and data exfiltration activities by analyzing behavior patterns of users and entities. It leverages advanced machine-learning algorithms to establish baseline

behaviors and identify anomalies that deviate from these norms, signaling potential security risks. UEBA provides actionable insights, enabling security teams to proactively respond to threats based on behavior analytics, improving an organization's security posture.

Stellar Cyber's Intrusion Detection System (or IDS) capabilities are designed to monitor network and system activities for malicious actions or policy violations. It employs advanced analytics to detect a wide range of threats-- from known malware to sophisticated zero-day attacks--by analyzing traffic patterns and signatures. The system prioritizes alerts to reduce noise, enabling security teams to focus on genuine threats and respond more effectively to incidents, thus enhancing the overall security posture of an organization.

12:00-13:00

Stellar Cyber's File Integrity Monitoring (or FIM) capabilities focus on protecting critical system and configuration files from unauthorized changes, which could indicate a cybersecurity threat. It continuously monitors and records changes to the file system, alerting security teams to any unauthorized modifications, deletions, or access that deviate from established security policies. This enables quick detection of potential security breaches, helping to maintain the integrity of the system and ensuring compliance with regulatory standards.

Operational Technology (or OT) Security, focuses on protecting systems that monitor and control physical devices, processes, and events in industrial environments, such as manufacturing plants, power grids, and transportation systems. It aims to secure the hardware and software that directly interacts with the physical world, safeguarding against cyber threats that could disrupt operational processes, cause physical damage, or endanger human safety.

13:00-14:00

By providing visibility across both IT and OT environments, Stellar Cyber ensures the integrity and availability of critical operational processes, enhancing the security posture against increasingly sophisticated cyber threats.

Stellar Cyber's Malware Sandbox is designed to safely execute and analyze suspicious files and URLs in an isolated environment to detect potential malware without risking the organization's network. By mimicking real user environments, it observes behaviors and outcomes of executed files to identify malicious activities and signatures. This process helps in enhancing threat detection and improving the organization's overall security posture by preventing malware infections before they can cause harm.

Stellar Cyber's Threat Intelligence Platform aggregates and analyzes threat data from various sources to provide actionable intelligence on potential cyber threats. It integrates this intelligence into security operations, enhancing detection capabilities and enabling proactive defense strategies.

14:00-15:00

By correlating real-time and historical data with threat intelligence, Stellar Cyber helps in identifying and prioritizing threats, ensuring timely and effective response to safeguard the organization's digital assets.

Stellar Cyber's Case Management functions streamline the investigation lifecycle by combining machine-learning Alert correlation, automated investigation processes, and intuitive aggregation workflows.

This system groups related Alerts into Cases for improved investigation outcomes, offers holistic views of threats through various visualizations, and optimizes investigation workflows allowing analysts to take bulk actions.

These capabilities significantly speed up the investigation and remediation of security threats. Additionally, the Case Activity and Evidence Locker features help manage the transfer of investigation knowledge among various teams and security stakeholders.

15:00-16:00
Case Management will be covered in depth in the *Working with Cases and Alerts* module of the SOC Analyst course. Stellar Cyber's Automated Response Actions work through its integrated Automation and Response capabilities. By utilizing pre-defined playbooks, the platform automates the response to detected threats, streamlining actions such as isolating endpoints, blocking IPs, or quarantining malicious files. This automation ensures consistent security outcomes, reduces the time to respond to incidents, and alleviates the manual workload on security teams. Additionally, actions can be taken manually and with automation at the Alert and Case level.

Stellar Cyber functions as a next-generation Security Information and Event Management (or SIEM) tool by integrating advanced analytics, machine learning, and comprehensive data collection from various sources to enhance threat detection, investigation, and response.

16:00-17:00
It automates the normalization and correlation of data across the network, cloud, and endpoints, providing actionable insights and real-time threat intelligence. This approach offers a unified, scalable platform for managing security events, reducing complexity, and improving the efficiency of security operations teams.

Let's take a moment to check your knowledge.

What does Stellar Cyber's Network Detection and Response, or NDR, capabilities utilize to monitor network traffic in real time?

A. Antivirus software
B. Firewalls
C. Advanced analytics and ML
D. Password protection mechanisms

The correct answer is C: Stellar Cyber's Network Detection and Response, or NDR, capabilities leverage advanced analytics and Machine Learning to monitor network traffic in real-time.

17:00-18:00
True or False? The Malware Sandbox detects malware infections by observing the behaviors of executed files in an environment isolated from your network.

The correct answer is True. Stellar Cyber's Malware Sandbox is designed to safely execute and analyze suspicious files and URLs in an isolated environment to detect potential malware without risking the organization's network.

What does Stellar Cyber's Intrusion Detection System (IDS) monitor for?

A. Network and system activities for malicious actions
B. Changes in employees work schedules
C. Unusual fluctuations in data storage usage
D. Email traffic volume and integrity

The correct answer is A. Stellar Cyber's IDS monitors for network and system activities for malicious actions or policy violations.

Stellar Cyber detects threats through advanced analytics and machine-learning algorithms, analyzing data across the network, endpoints, and cloud to identify anomalies and indicators of compromise.

18:00-18:30
It integrates diverse security data sources, enhancing detection capabilities, and enabling a comprehensive view of a threat landscape. The platform automates Alert correlation, reducing false positives and focusing on high-fidelity threats, thereby streamlining the response process and improving overall security posture.

The lesson, *Components of Open XDR*, is complete. You should now be familiar with the powerful integrations and native components Stellar Cyber uses to protect the network. Thank you.

L06 - MODULAR SENSOR FUNCTIONS

L06A-PRES: Modular Sensor Functions (09m:35s)

00:00-01:00
The next lesson in the *Open XDR Explained* module is *Modular Sensor Functions. Modular Sensor Functions*. In this lesson, you will learn about Modular Sensors, one of the three main types of sensors in the suite of sensors in the Stellar Cyber arsenal.

A modular sensor in Stellar Cyber refers to a flexible, scalable component of the Stellar Cyber security platform designed to collect, normalize, and analyze data from various sources across the network.

These sensors can be deployed across different parts of an organization's infrastructure to monitor network traffic, log files, and other security-relevant data, adapting to specific security needs and scaling as necessary. This modular approach allows for tailored deployment strategies that match the unique environments and security challenges of different organizations.

01:00-02:00
There are three main types of sensors in the suite of sensors in the Stellar Cyber arsenal:

- Linux Server Sensors
- Windows Server Sensors, and
- Modular Sensors

The following are some of the functions achieved by using modular sensors.

Log forwarding in a SIEM is essential for centralizing security data from across an organization's infrastructure, enabling real-time analysis and monitoring of security events. It ensures that logs from various sources, such as servers, network devices, and applications, are collected in one place; enhancing the ability to detect, investigate, and respond to potential security threats efficiently. This process is critical for maintaining a comprehensive security posture and facilitating compliance with regulatory requirements by ensuring all relevant data is available for analysis and reporting by providing a unified view of security-related data.

Stellar Cyber analyzes **Network Traffic Metadata** by collecting, normalizing, and analyzing data from network flows and packets. It employs advanced machine learning and behavior analysis to detect anomalies, threats, and patterns indicative of cyberattacks**.**

02:00-03:00
This process enables detailed visibility into network activities, facilitating the identification of malicious behavior and aiding in the swift response to potential security incidents.

Deep packet inspection (or DPI) is a technology used to analyze and investigate network traffic at the packet level. DPI can identify and classify protocols, applications, and even specific content within the packets, allowing for advanced network monitoring and security analysis.

DPI aims to manage network traffic and analyze it for a variety of security and administrative purposes, including to detect and prevent cyber threats. It enables a more sophisticated level of monitoring of data packets, allowing for detailed insights into network activities.

03:00-04:00
Stellar Cyber's **Intrusion Detection System (or IDS)** is designed to monitor network and system activities for malicious actions or policy violations.

Utilizing advanced analytics, it detects a wide range of threats, from known malware to sophisticated zero-day attacks, by analyzing traffic patterns and signatures. This system prioritizes alerts to reduce noise, enabling security teams to focus on genuine threats and respond more effectively to incidents, thus enhancing the overall security posture of an organization.

Let's take a moment to check your knowledge.

True or False? Stellar Cyber's arsenal includes only Linux Server Sensors and Windows Server Sensors.

The correct answer is False. There are three main types of sensors in the suite of sensors in the Stellar Cyber arsenal: Linux Server Sensors, Windows Server Sensors, and Modular Sensors.

04:00-05:00
What is the primary purpose of Stellar Cyber Modular Sensors?

A. To collect and store data
B. To collect, analyze, and normalize data
C. To serve as a firewall
D. To store data

The correct answer is B: Stellar Cyber Modular Sensors are designed to collect, normalize, and analyze data from various sources across the network.

True or False? Stellar Cyber Modular Sensors can adapt to specific security needs and scale as necessary.

The correct answer is True. Stellar Cyber Modular Sensors can be deployed across different parts of an organization's infrastructure to monitor network traffic, log files, and other security-relevant data, adapting to specific security needs and scaling as necessary.

05:00-06:00
True or False? Stellar Cyber's Intrusion Detection System (IDS) can only detect known malware.

The correct answer is False. Stellar Cyber's Intrusion Detection System detects a wide range of threats, from known malware to sophisticated zero-day attacks.

Stellar Cyber's **Malware Sandbox** is designed to safely execute and analyze suspicious files and URLs in an isolated environment to detect potential malware without risking the organization's network.

By mimicking real user environments, it observes behaviors and outcomes of executed files to identify malicious activities and signatures. This process helps in enhancing threat detection and improving the organization's overall security posture by preventing malware infections before they can cause harm.

In a SIEM system, **log parsing** is the process of breaking down log files from various sources into a structured format that can be analyzed.

06:00-07:00
This involves extracting meaningful information from raw logs to identify security events or incidents. The parsed data is then normalized, allowing the SIEM to correlate events across different systems and detect potential threats or vulnerabilities efficiently.

Data normalization in a SIEM is crucial because it standardizes data from diverse sources into a common format facilitating efficient analysis, correlation, and alerting. It enables the SIEM to accurately compare and link related security events across different systems and applications, improving threat detection and incident response. Normalization simplifies data management, enhances visibility into security postures, and supports compliance requirements.

Interflow Records are pivotal in Stellar Cyber for aggregating and normalizing data from diverse security tools and network elements into a standardized format. This uniformity allows for more effective analysis, correlation, and detection of cyber security threats across an organization's digital landscape.

07:00-08:00
By leveraging Interflow Records, Stellar Cyber enhances visibility, streamlines threat detection processes, and facilitates rapid response to identified security incidents.

API connectors in Stellar Cyber facilitate integration with various external security tools and data sources,enabling seamless data ingestion and interaction with the platform. They allow Stellar Cyber to extend its threat detection and response capabilities by leveraging external intelligence and services. Through these connectors, users can automate the collection, normalization, and analysis of data from disparate systems, enhancing the efficiency and effectiveness of security operations.

Let's take a moment to check your knowledge.

What is the role of log parsing in a SIEM system?
A) To create new log files
B) To break down log files into a structured format for analysis
C) To archive old log files
D) To forward log files to network administrators

08:00-09:00
The correct answer is B. In a SIEM system, log parsing is the process of breaking down log files from various sources into a structured format that can be analyzed.

Why is data normalization crucial in a SIEM?
A) To reduce the size of the data
B) To remove errors from the data

C) To standardize data for efficient analysis and correlation
D) To encrypt the data

The correct answer is C. Data normalization in a SIEM is crucial because it standardizes data from diverse sources into a common format, facilitating efficient analysis, correlation, and alerting.

What are Interflow Records used for in Stellar Cyber?

A) To create new security policies
B) To document security incidents
C) To serve as user manuals for the security platform
D) To aggregate and normalize data into a standardized format

The correct answer is D. Interflow Records are pivotal in Stellar Cyber for aggregating and normalizing data from diverse security tools and network elements into a standardized format.

09:00-09:35

What is the significance of connectors in Stellar Cyber?

A) They provide physical connections to network devices.
B) They enable the platform to extend threat detection capabilities by leveraging external intelligence.
C) They are used for charging mobile devices.
D) They serve as decorative elements in the software interface.

The correct answer is B. Connectors allow Stellar Cyber to extend its threat detection and response capabilities by leveraging external intelligence and services.

The lesson, modular sensor functions, is complete. You should now be familiar with Modular Sensors, one of the three main types of sensors in the suite of sensors in the Stellar Cyber arsenal. Thank you!

L07 - THREAT DETECTION ENRICHMENT

L07A-PRES: Threat Detection Enrichment (08m:08s)

00:00-01:00

The next lesson in the Open XDR Explained module is *Threat Detection Enrichment*. Threat Detection Enrichment. In this lesson, you will learn how Enrichment in the Stellar Cyber User Interface helps analysts to understand the nature and severity of cybersecurity incidents and respond quickly and appropriately.

Steller Cyber helps analysts get answers to questions more quickly by providing relevant contextual information about the Cases, Alerts, and events they are seeing in the platform. Stellar Cyber integrates the MITRE ATT&CK framework into its Alert and Case display system, showcasing the attack stage, tactics, and techniques associated with each security Alert or Case. This approach helps

security analysts quickly understand the context and methodology of detected threats, aligning them with known adversary behaviors.

01:00-02:00
By categorizing and displaying this information, Stellar Cyber facilitates a more informed and efficient response, enabling teams to prioritize their efforts based on the sophistication and potential impact of the attack strategies used. In the User Interface, it shows the stage of the attack kill chain, the MITRE tactic and technique. These appear as hyperlinks that take you to the MITRE website. This greatly helps facilitate understanding of the nature of the event itself. As we mentioned before, no analyst can know every type of attack and the significance of each. Having this display in the UI allows analysts to determine how critical the event is, even if it is a concept that has never been encountered by the analyst. Additionally, the UI gives a clear description of what the alert signifies. Here we have a potential Mimikatz memory dump on ot-workstation.incident.ad (IP 10.33.4.135)...

02:00-03:00
...initiated from C/Users/Administrator/Downloads/mimikats_trunk/x64/mimikatz.exe with access_mask of 0x1010 was detected. This lets us know what this alert means. What is OS Credential Dumping? Clicking the MITRE links in the UI gets us critical information about what credential dumping means. This view is also enriched with information about the Case or Alert score, how critical the event is, and information about how serious Stellar Cyber considers the event to be. This allows for complex analysis and identification of related events. Finally, Stellar Cyber's data framework requires certain fields to help identify the data source, category, and type of event. You can see the impacted users and assets. Also visible are the processes being spawned by the attacker. The chronological timeline paints the picture of what exactly is happening on your network.

03:00-04:00
Detailed description enrichments help analysts understand exactly what is happening on the network or assets involved. In this event we can see that a potentially malicious Windows process was created on the host (with the logged in username SYSTEM. This detection was triggered by a rule with the following description: "Multiple system commands have been executed in a short period of time. These commands are often used by malicious actors to recon a target once they get access." If a cyber attacker runs a process as SYSTEM, it means they have obtained the highest level of privileges on a Windows system, equivalent to root access on UNIX/Linux systems. This level of access allows the attacker to execute commands and modify system settings without any restrictions, potentially leading to complete control over the system, data theft, system damage, or deployment of malware. Achieving SYSTEM privileges is often a critical step in a sophisticated attack. It enables broad exploitation and persistence mechanisms within the targeted environment.

04:00-05:00
Mimikatz is a hacking tool that was originally developed to demonstrate Windows security vulnerabilities. It is widely used by attackers and cyber security professionals to extract plaintext passwords, hash, PIN codes, and Kerberos tickets from memory. Mimikatz can also perform pass-the-hash, pass-the-ticket, or build Golden tickets, making it a versatile tool for anyone looking to test or bypass Windows security features. Here we can see that with automated correlation and

threat event enrichment, it is very clear, in most cases, what exactly is occurring. In this example it is abundantly clear that this host needs to be isolated in order to prevent lateral movement on your network. We can assess the criticality of the event with just this one Alert. All of these functions form a comprehensive suite of protection for your network. When malicious activity does occur on your network, the user interface presents each action taken by the attacker in its proper context.

05:00-06:00
Alerts are described in plain English and automatically include critical contexts like threat intelligence and asset criticality. This data enrichment happens automatically, so new or junior analysts can make more accurate assessments of the nature of the attack and respond appropriately. No team wants critical incidents dismissed because the activity is poorly or incorrectly understood. Not catching bad actions early creates lots of hours of remediation as well as potential financial or reputational damage down the road. Another way Stellar Cyber enriches threat info is it identifies key characteristics by comparing fields and values to identify duplicates. This helps eliminate noise and reduce Alert fatigue. Also, Stellar Cyber standardizes incoming data sources to a common data model, mapping values from the raw data into standardized fields.

06:00-07:00
In Stellar Cyber, enriched fields typically include threat intelligence indicators such as IP reputation, domain risk scores, and malware signatures. These fields add context to alerts by associating them with known threats, facilitating quicker and more informed decision-making during investigations. This enrichment helps analysts to prioritize their efforts based on the severity and credibility of the threat indicators involved.

Let's take a moment to check your knowledge.

True or False? The Stellar Cyber user interface incorporates the MITRE ATT&CK framework into its Alert and Case display system.

The correct answer is True. Stellar Cyber integrates the MITRE ATT&CK framework into its Alert and Case display system, showcasing the attack stage, tactics, and techniques associated with each security Alert or Case.

07:00-08:08
True or False? The Stellar Cyber user interface provides hyperlinks to the MITRE website to help analysts understand the nature of cybersecurity events.

The correct answer is True. The correct answer is True. There are hyperlinks in the Stellar Cyber user interface that link directly to the MITRE website, helping facilitate understanding of the nature of the event.

What is Mimikatz commonly used for?

A. Enhancing Windows security
B. Backing up user data

C. Installing Windows updates
D. Extracting plaintext passwords from memory

The correct answer is D. Mimikatz is widely used by attackers and cybersecurity professionals to extract plaintext passwords (as well as hash, PIN codes, and Kerberos tickets) from memory.

The lesson, Threat Detection Enrichment, is complete. You should now be familiar with how Enrichment in the Stellar Cyber User Interface helps analysts understand the nature and severity of cybersecurity incidents and respond quickly and appropriately. Thank you!

L08 - RESPONSE ACTIONS

L08A-PRES: Response Actions (05m:15s)

00:00-01:00
The next lesson in the Open XDR Explained module is response actions.

Response actions. In this lesson, you will learn about the analyst-initiated and automated response actions available in the Stellar Cyber platform.

Now let's talk a little about the analyst-initiated and automated response actions available in the Stellar Cyber Platform.

Stellar Cyber enables analysts to take immediate actions on users or assets when an event occurs. API connections to security platforms allow for automated or analyst-initiated actions against threats.

Stellar Cyber has native response actions built into its platform. Users can respond back through a connector such as an EDR or identity provider in one of two ways. First, mid-investigation, the full response action menu is available, so a user does not have to move to another screen.

01:00-02:00
Second, users can leverage automated playbooks to take certain actions based on any criteria to automate as many common workflows as possible.

Stellar Cyber offers a range of response actions to manage and mitigate threats effectively. These actions can be configured and managed through **Respond | Actions** page and include the ability to take direct actions on events based on the deployment configuration. The system allows for automated threat hunting to trigger reports, which can be defined using dashboards.

Here are the types of response actions available in Stellar Cyber: firewall actions, user actions, script actions, endpoint actions, email actions, security switch actions, and webhook actions. These actions

can be submitted from various pages such as event details or automated threat hunting, providing a comprehensive approach to responding and to managing security events.

02:00-03:00
We'll explore how to use these functions in the *Intro to the Stellar Cyber User Interface* module.

These immediate actions could be the difference between a minor inconvenience and a costly breach. We'll see more of this functionality in the *User Interface* module of this course. Understanding what types of data you'll be seeing is a very important first step learning how to investigate and respond in Stellar Cyber.

To better understand the various alerts and security events you'll see in the user interface, it helps to have a working understanding of the architecture of the platform and the data that powers it. Some examples of response actions that can be taken at the Alert level are Blocking an IP, disabling an account, quarantining a threat, initiating a scan, adding an Alert filter, triggering an email to be sent to security stakeholders or employees, running a script, shutting down hosts.

03:00-04:00
Response actions can be initiated from the Case level as well. Examples are blocking an IP, disabling an account, running a script, containing a host, disconnecting a host, initiating a vulnerability scan, restarting a machine, shutting down hosts, killing a threat, quarantining a threat, remediating threats, remediation of email. Remediation email message actions are either none, delete, or quarantine. So you can quarantine an email and then search for it in other places if it contains a threat.

Let's take a moment to check your knowledge.

True or False: The Stellar Cyber platform allows for both automated or analyst-initiated actions against threats.

04:00-05:15
The correct answer is true. API connections to security platforms allow for automated or analyst-initiated actions against threats.

What are the remediation email message actions available in Stellar Cyber?
A) Forward, reply, or archive
B) None, delete, or quarantine
C) Mark as read, save to drafts, or flag
D) Encrypt, bounce back, or mark as spam

The correct answer is B. Remediation email message actions are either none, delete, or quarantine so that you can quarantine an email and then search for it in other places if it contains a threat.

Where can response actions be submitted from in Stellar Cyber?
A) Only from the event details page

B) Only from the automated threat hunting page
C) Various pages, including event details and automated threat hunting
D) Solely from the Respond actions page

The correct answer is C: Response actions can be submitted from various pages such as event details or automated threat hunting, providing a comprehensive approach to responding and to managing security events.

The lesson, *Response Actions*, is complete. You should now be familiar with the analyst-initiated and automated response actions available in the Stellar Cyber platform.

Thank you!

L09 - KEY DIFFERENTIATORS

L09A-PRES: Key Differentiators (05m:45s)

00:00-01:00
The next lesson in the Open XDR Explained module is *Key Differentiators of Open XDR*.

Key Differentiators of Open XDR. In this lesson you'll learn how Stellar Cyber's Open Extended Detection and Response fits into the information security ecosystem.

We're diving into the world of Stellar Cyber's Open Extended Detection and Response, and its unique role within the information security ecosystem.

In an era where cyber threats are evolving at an unprecedented pace, traditional security measures are no longer sufficient.

Organizations are faced with the daunting task of integrating disparate security tools, managing overwhelming volumes of alerts, and deciphering complex threats with limited resources. This is where Stellar Cyber's Open XDR comes into play, offering a revolutionary solution that stands out in the cybersecurity landscape.

01:00-02:00
Stellar Cyber's Open XDR is designed to bridge the gaps between isolated security tools, providing a unified platform that not only collects and correlates data across various security sources but also applies advanced analytics to detect, investigate, and respond to threats more efficiently and effectively. Unlike traditional security measures, Open XDR transcends the limitations of siloed approaches, enabling a more comprehensive and proactive defense mechanism.

But what sets Stellar Cyber's Open XDR apart in the information security ecosystem? It's the platform's openness and flexibility. Open XDR integrates seamlessly with existing security tools, enhancing their capabilities without the need for replacing them.

This not only optimizes security investments but also ensures a more robust and adaptive security posture that can evolve with the changing threat landscape.

02:00-03:00
Stellar Cyber empowers security teams with deeper insights, streamlined operations, and a more resilient security framework.

In this lesson, we'll delve into its key differentiators, and the tangible benefits it brings to organizations striving to navigate the complexities of today's cybersecurity challenges.

Stellar Cyber is vendor agnostic. Unlike proprietary XDR solutions, Open XDR works across different vendors' products, offering flexibility in security stack composition.

Integration and Interoperability: It facilitates seamless integration and interoperability between different security technologies and platforms.

Customization and Control: It offers organizations the ability to customize their security setup by choosing best-of-breed components.

Unified Management: It provides a centralized management console for monitoring and responding to threats across diverse security solutions.

03:00-04:00
Data Source Inclusivity: It's capable of ingesting data from a wide range of sources, including third-party tools and cloud sources.

Let's take a moment to check your knowledge.

What does "vendor agnostic" mean in the context of Stellar Cyber Open XDR?

A. It works only with one specific vendor's products
B. It doesn't work with any vendor's products
C. It works with various security vendors' products
D. It is not related to vendors

The correct answer is C. In the context of Stellar Cyber Open XDR, "vendor agnostic" means that it works with various security vendors' products.

Now let's get back to more key differentiators of Stellar Cyber Open XDR.

Cost-Effectiveness: It reduces the need for a complete overhaul of existing security infrastructure, protecting previous investments.

Scalability: It easily scales with organizational needs without being locked into a single vendor's ecosystem.

04:00-05:00
Community-Driven Enhancements: It benefits from community-driven developments and shared threat intelligence.

Flexibility in Response Actions: It enables flexible response actions utilizing a variety of tools from different vendors within the security stack.

Comprehensive Coverage: It aims to provide comprehensive threat detection and response across the entire technology stack, including endpoints, networks, and cloud.

Artificial Intelligence and Machine Learning: It harnesses the power of AI and machine learning from diverse systems to enhance threat detection and predictive analytics.

Real-Time Collaboration: It facilitates real-time collaboration and information sharing between different security components for faster threat resolution.

Challenges and Considerations: It addresses compatibility and complexity in managing disparate systems while ensuring cohesive security operations.

05:00-05:45
Let's take a moment to check your knowledge.

What is the significance of comprehensive coverage in Stellar Cyber Open XDR?

A. It covers potential security threats
B. It eliminates the need for AI and machine learning
C. It provides a unified management console
D. It focuses only on endpoint security

The correct answer is A. The significance of comprehensive coverage in Stellar Cyber Open XDR is that it covers potential security threats.

The lesson, *Key Differentiators of Open XDR*, is complete. You should now be familiar with how Stellar Cyber Open XDR fits into the information security ecosystem. Thank you!

L10 - SUMMARY

L10A-PRES: Open XDR Explained Summary (02m:20s)

00:00-01:00
In the *Open XDR Explained* module, we looked at the many layers of a security solution designed to redefine the cyber security landscape.

From understanding the foundational principles of Open XDR to unveiling its multifaceted benefits, features, components, and the strategic advantage it offers through Modular Sensor functions, threat detection enrichment, and responsive actions, we hope you have a better understanding of where Stellar Cyber fits in the security ecosystem and its benefits. What makes Open XDR a standout in the cybersecurity arena is its ability to offer a cohesive, integrated approach to detection and response that is reactive, proactive, and predictive.

The key differentiators of Stellar Cyber's platform--including its open ecosystem, scalability, and intelligent automation--empower organizations to combat the current spectrum of cyber threats...

01:00-02:00
...as well as adapt to emerging challenges with agility and confidence.

The benefits of Open XDR are clear: **enhanced detection capabilities**, **streamlined response actions**, and a **unified security posture** that leverages the full potential of your existing security investments. With the Stellar Cyber Open XDR platform, organizations can achieve a higher level of security intelligence, operational efficiency, and resilience against cyber threats.

As we conclude, it's important to reflect on the significance of embracing an Open XDR approach in today's complex and evolving cyber threat landscape. The journey towards a more secure digital environment is ongoing, and tools like Stellar Cyber's Open XDR are at the forefront of this transformation--offering sophisticated, integrated, and comprehensive cyber security solutions.

02:00-02:20
Thank you for joining us to discover how Stellar Cyber Open XDR can be the linchpin in your organization's cybersecurity strategy, enabling a more secure, responsive, and resilient future.